Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)

On

Engineering Scalable Applications with Optimised Web Performance

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Introduction

This research focuses on advanced web performance optimisation in React applications, studying three core engineering patterns and their measurable impact.

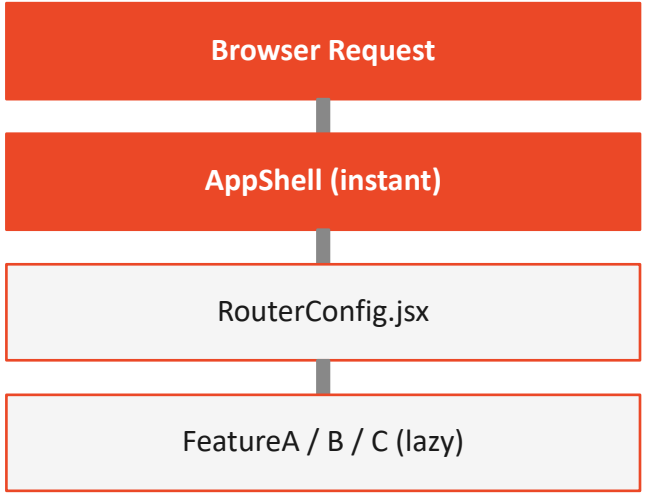
Key Areas of Study:

- AppShell Architecture — lightweight shell for instant first paint with lazy-loaded features
- Sequential vs. Parallel Data Fetching — comparing cumulative vs. concurrent API strategies
- Bundle Optimisation & Import Hygiene — tree-shaking via direct imports over barrel imports
- Performance measurement using DevTools and real browser benchmarks
- Practical showcase built with React 18 + Vite 4 to demonstrate findings

AppShell Architecture & Lazy Loading

Pattern 01 — Instant first paint by separating the shell from data-driven content:

- AppShell.jsx loads as a minimal skeleton — renders in under 200ms
- RouterConfig.jsx handles client-side routing with lazy boundaries
- LazyLoader.jsx wraps each React.lazy() import with a Suspense fallback
- FeatureA, FeatureB, FeatureC are independent code-split chunks
- Unused routes are never downloaded until the user navigates to them
- Result: ~62% improvement in initial load time vs. eager loading



React.lazy() + Suspense

Sequential vs. Parallel Data Fetching

Pattern 02 — Independent API calls should always fire concurrently:

✗ Sequential Fetching (~1050ms)

- await fetchUser() → 300ms
- await fetchPosts() → 400ms
- await fetchComments() → 350ms
- **Total: 300 + 400 + 350 = 1050ms**

✓ Parallel Fetching (~400ms)

- Promise.all([fetchUser(), fetchPosts(), fetchComments()])
- All three requests fire simultaneously
- Resolves when the slowest completes: ~400ms
- **Total: max(300, 400, 350) = 400ms (2.6× faster)**

Key insight: use Promise.all() for any independent async operations — never chain awaits unnecessarily.

Bundle Optimization & Technology Used

Import Hygiene — Pattern 03:

- Barrel imports (index.js re-exports) pull entire modules into the bundle
- Direct imports enable Vite's tree-shaking — dead code is eliminated
- Bad: import { Button } from '@ui/components' (+80–200KB)
- Good: import Button from '@ui/components/Button' (−60–80%)
- Combined with lazy loading: overall bundle size reduced by ~70%

Tech Stack

React 18

Datadog

Jest

LogRocket

Also: Vite 4 · React Router · ES6 Modules · Vanilla CSS

Results & Key Takeaways

~62%	2.6×	~70%
Faster Initial Load	**Faster Data Fetch**	**Smaller Bundle**
AppShell + lazy loading	Parallel over sequential	Direct imports + Vite tree-shaking

Key Takeaways:

- AppShell is the foundation — skeleton loads instantly, features stream in on demand
- Always use Promise.all() for independent API calls, never chain awaits
- Barrel imports are a hidden bundle tax — replace with direct, tree-shakeable imports
- Measure before optimizing — use DevTools Performance tab to identify real bottlenecks

Thank You

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